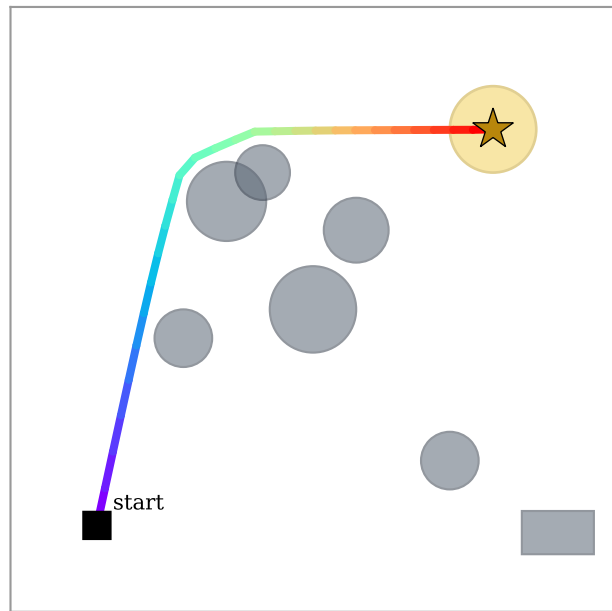
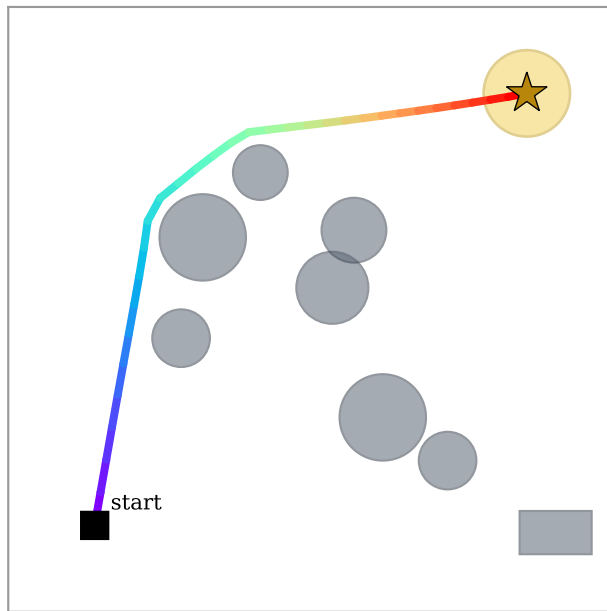


# TeLoGraF flow-matching trajectories on a cluttered map (trajectory colour = time)

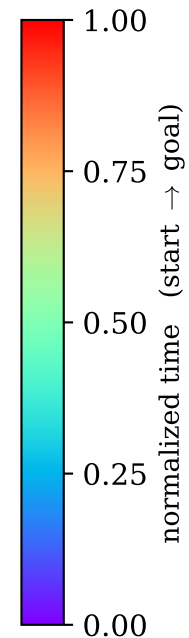
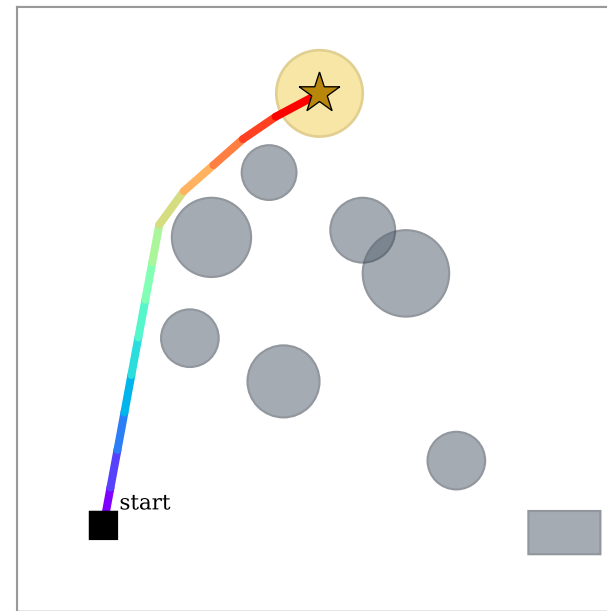
(a) reach\_within\_T TeLoGraF (flow + STL CG)



(b) reach\_avoid TeLoGraF (flow + STL CG)



(c) reach\_goal\_north TeLoGraF (flow + STL CG)



NL: "Reach the goal within 12 time units while always keeping safe from every obstacle."

STL:  $F_{[0,12]}(\text{reach}) \wedge G \neg \text{unsafe} \quad \rho(G \neg \text{unsafe}) = +0.20 \text{ m}$

NL: "Always avoid all obstacles and eventually, within the time bound, reach the goal."

STL:  $F_{[20,55]}(\text{reach}) \wedge G \neg \text{unsafe} \quad \rho(G \neg \text{unsafe}) = +0.20 \text{ m}$

NL: "Keep safe from every obstacle and reach the north goal within the time bound."

STL:  $F_{[20,55]}(\text{reach}) \wedge G \neg \text{unsafe} \quad \rho(G \neg \text{unsafe}) = +0.20 \text{ m}$